MATH 202A: Introduction to Topology and Analysis

By William Nguyen

Fall 2026

Contents

10/30/26 Lecture 27: Absolute Continuity and the Lebesgue FTC 1
Definition 27.2: Absolute Continuity 1
Theorem 27.6: Fundamental Theorem of Calculus for Lebesgue Integrals 3
Definition 27.7: Discrete and Continuous Measures 3

10/30/26 Lecture 27: Absolute Continuity and the Lebesgue FTC

Recall: A function $F : \mathbb{R} \to \mathbb{R}$ is BV iff it can be written as a difference of bounded increasing functions (Theorem 26.3), and

$$\text{NBV} = \{F \in \text{BV} : F \text{ is right continuous and } F(-\infty) = 0\}$$

Finite signed Borel measures on $\mathbb{R}$ correspond to NBV functions (Theorem 26.6).

Proposition 27.1

If $F \in \text{NBV}$, then $F' \in L^1$. Moreover, the LS measure m_F satisfies

$$m_F \perp \lambda \iff F' = 0 \text{ a.e.}$$

and

$$m_F \ll \lambda \iff F(x) = \int_{-\infty}^{x} F'(y)\, dy$$

Definition 27.2: Absolute Continuity

A function $F : [a, b] \to \mathbb{R}$ is *absolutely continuous* if for every $\epsilon > 0$ there is $\delta > 0$ such that, given finitely many disjoint intervals $(a_1, b_1), \ldots, (a_N, b_N)$ in $[a, b]$ with

$$\sum_{j=1}^{N} (b_j - a_j) < \delta$$

then

$$\sum_{j=1}^{N} |F(b_j) - F(a_j)| < \epsilon$$

Example: Bounded Derivative

If F is differentiable and $|F'| \leq M$, then by the mean value theorem,

$$|F(b_j) - F(a_j)| \leq |F'(c_j)|\,(b_j - a_j) \leq M(b_j - a_j)$$

for some $c_j \in (a_j, b_j)$. Given ϵ, make $\delta = \frac{\epsilon}{M}$.

Example: The Cantor Function

An NBV function that is not AC is the Cantor function F_c, since $F_c' = 0$ a.e.

Proposition 27.3

If $F \in \text{NBV}$, then F is absolutely continuous on $\mathbb{R}$ iff $m_F \ll \lambda$.

Proof: If $m_F \ll \lambda$, for every ϵ we can find δ so that if $\lambda(A) < \delta$, then $|m_F|(A) < \epsilon$. So if $\lambda\left(\bigcup_j (a_j, b_j)\right) < \delta$,

$$\sum_j |F(b_j) - F(a_j)| = \sum_j |m_F((a_j, b_j])| \le |m_F|\left(\bigcup_j (a_j, b_j]\right) < \epsilon$$

Conversely, let A such that $\lambda(A) = 0$. Let $\epsilon > 0$, and pick the $\delta > 0$ that comes from absolute continuity of F. By outer regularity, we can find open sets $U_1 \supseteq U_2 \supseteq \cdots \supseteq A$ so that $|m_F|(U_k) \to |m_F|(A)$. Also, we can find V open so that $\lambda(V) < \delta$ and $A \subseteq V$, so $\lambda(U_n \cap V) < \delta$.

Write $U_n \cap V = \bigcup_{j=1}^{\infty} (a_{nj}, b_{nj})$ as a disjoint union of open intervals. By absolute continuity, for every finite N,

$$\sum_{j=1}^{N} |m_F((a_{nj}, b_{nj}))| = \sum_{j=1}^{N} |F(b_{nj}) - F(a_{nj})| < \epsilon$$

Taking $N \to \infty$, $|m_F(U_n \cap V)| \le \epsilon$, so letting $n \to \infty$, $|m_F(A)| \le \epsilon$. Taking $\epsilon \to 0$, we get $m_F(A) = 0$. ■

Corollary 27.4

If $f \in L^1$, then $F(x) = \int_{-\infty}^{x} f(y)\,dy$ is NBV and absolutely continuous, with $F' = f$ a.e.

Conversely, if $F \in$ NBV and is AC, then $F' \in L^1$ and $F(x) = \int_{-\infty}^{x} f(y)\,dy$ where $f = F'$ a.e.

Proposition 27.5

If F is AC on $[a, b]$, then it is BV on $[a, b]$.

Proof: Choose δ so that if $\sum_j (b_j - a_j) < \delta$, then

$$\sum_j |F(b_j) - F(a_j)| < 1 \quad (\epsilon = 1)$$

Divide $[a, b]$ into N pieces $I_1, \ldots, I_N$ with width less than δ.

We want to show that, given $a = x_0 < \cdots < x_n = b$,

$$\sum_{j=1}^{n} |F(x_j) - F(x_{j-1})| < L$$

for any partition. We can always add the endpoints of the δ-intervals to the partition.

Group the set of points into intervals: since $|I_k| < \delta$, set N groups, and for every group G,

$$\sum_{j \in G} |F(x_j) - F(x_{j-1})| < 1$$

so in total,

$$\sum_{j=1}^{n} |F(x_j) - F(x_{j-1})| < N$$

■

Theorem 27.6: Fundamental Theorem of Calculus for Lebesgue Integrals

For $F : [a, b] \to \mathbb{R}$ (on a finite interval), the following are equivalent:

1. F is AC.
2. $F(x) - F(a) = \int_a^x f(y)\, dy$ for some $f \in L^1([a, b])$.
3. F is differentiable a.e. with $F' \in L^1([a, b])$ and $F(x) - F(a) = \int_a^x F'(y)\, dy$.

Proof: $(1 \Rightarrow 3)$ Let $G : \mathbb{R} \to \mathbb{R}$,

$$G(x) = \begin{cases} F(b) - F(a) & x \geq b \\ F(x) - F(a) & x \in [a, b) \\ 0 & x < a \end{cases}$$

G is NBV. This implies it is differentiable a.e., and

$$F(x) - F(a) = \int_{-\infty}^{x} G'(y)\, dy = \int_a^x F'(y)\, dy$$

which gives 3.

$(3 \Rightarrow 2)$ Immediate.

$(2 \Rightarrow 1)$ Extend f by 0 outside $[a, b]$ and apply Corollary 27.4.

■

Definition 27.7: Discrete and Continuous Measures

A Borel measure is *discrete* if it can be written as

$$\sum_{j=1}^{\infty} a_j \delta_{x_j}, \quad x_j \in \mathbb{R}^d, \ a_j \in \mathbb{R}$$

A measure μ is *continuous* if $\mu(\{x\}) = 0$ for all x.

Proposition 27.8

Let μ be a finite positive Borel measure. Then

$$\mu = \underbrace{\mu_d}_{\text{discrete}} + \underbrace{\mu_{ac}}_{\text{AC}} + \underbrace{\mu_{sc}}_{\text{singular continuous}}$$

Proof: Let $A = \{x : \mu(\{x\}) > 0\}$. Then

$$\{x : \mu(\{x\}) > 1/k\} \text{ is finite for every } k$$

because μ is finite, so A is countable. Then $\mu|_A$ is discrete and $\mu|_{A^c}$ is continuous.

Now, apply the Lebesgue decomposition to $\mu|_{A^c}$.

■